

Advanced Machine Learning

DATA 642

Lecture 3 — Linear Programming

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth

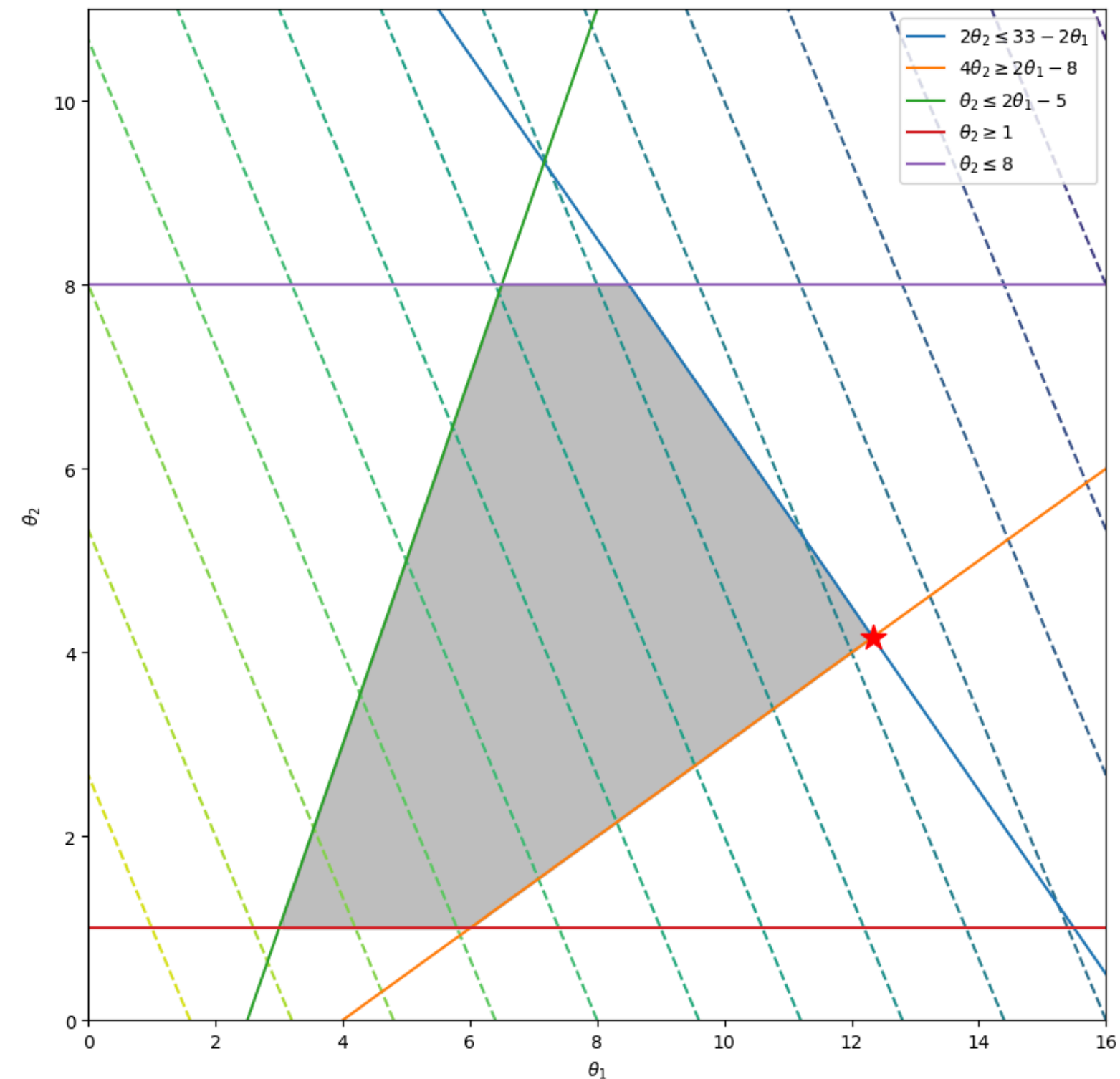
Linear Programming

Linear programming in machine learning is about finding the best possible solution to a problem while adhering to a set of linear constraints.

- **Optimization of Linear Models:** Linear programming provides a powerful framework for optimizing linear models, which are fundamental in many machine learning algorithms. It allows us to efficiently find the optimal parameters of linear models by minimizing or maximizing an objective function subject to linear constraints.
- **Constraint Handling:** In machine learning, it's common to have constraints on variables or parameters. Linear programming enables us to handle these constraints effectively, ensuring that the solutions obtained are feasible and satisfy the given constraints. This is crucial for modeling real-world problems where certain conditions or limitations must be met.
- **Feature Selection and Dimensionality Reduction:** Linear programming techniques can be applied to feature selection and dimensionality reduction tasks in machine learning. By formulating these problems as linear optimization tasks, we can efficiently identify the most relevant features or reduce the dimensionality of the data while preserving important information.

This illustration depicts a contour plot with dashed lines representing level curves. Five inequality lines are plotted with different slopes and intercepts, each represented by a dashed line and labeled with its corresponding inequality equation. Additionally, there are markers indicating specific points on the plot, and a shaded region is highlighted between certain curves.

$$\begin{array}{ll}
 \underset{\theta}{\text{minimize}} & -5\theta_1 - 3\theta_2 \\
 \text{subject to} & 2\theta_1 + 2\theta_2 \leq 33 \\
 & 2\theta_1 - 4\theta_2 \leq 8 \\
 & -2\theta_1 + \theta_2 \leq -5 \\
 & -\theta_2 \leq -1 \\
 & \theta_2 \leq 8
 \end{array}$$



The mathematical setup

Consider the special case when all the preceding functions are linear, i.e.,

$$\min_{\boldsymbol{\theta} \in \mathbb{R}^d} c^T \boldsymbol{\theta} \quad \text{subject to} \quad A\boldsymbol{\theta} \leq b \quad (1)$$

where $A \in \mathbb{R}^{m \times d}$ and $b \in \mathbb{R}^m$. This is known as a linear program. It has d variables and m linear constraints.

The Lagrangian is given by

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\lambda}) = c^T \boldsymbol{\theta} + \boldsymbol{\lambda}^T (A\boldsymbol{\theta} - b) \quad (2)$$

where $\boldsymbol{\lambda} \in \mathbb{R}^m$ is the vector of non-negative Lagrange multipliers.

Rearranging the terms corresponding to $\boldsymbol{\theta}$ yields

$$\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\lambda}) = (c + A^T \boldsymbol{\lambda})^T \boldsymbol{\theta} - \boldsymbol{\lambda}^T b : \quad (3)$$

Taking the derivative of $\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\lambda})$ with respect to $\boldsymbol{\theta}$ and setting it to zero gives us

$$c + A^T \boldsymbol{\lambda} = 0 : \quad (4)$$

Therefore, the dual Lagrangian is $D(\lambda) = -\boldsymbol{\lambda}^T \mathbf{b}$. Recall we would like to maximize $D(\boldsymbol{\lambda})$. In addition to the constraint due to the derivative of $\mathcal{L}(\boldsymbol{\theta}, \boldsymbol{\lambda})$ being zero, we also have the fact that $\boldsymbol{\lambda} \geq 0$, resulting in the following dual optimization problem:

$$\max_{\boldsymbol{\lambda} \in \mathbb{R}^m} -\mathbf{b}^T \boldsymbol{\lambda} \quad \text{subject to} \quad \mathbf{c} + \mathbf{A}^T \boldsymbol{\lambda} = 0, \quad \boldsymbol{\lambda} \geq 0 \quad (5)$$

This is also a linear program, but with m variables.

Note that we have the choice of solving the primal (1) or the dual program (5) depending on whether m or d is larger. Recall that d is the number of variables and m is the number of constraints in the primal linear program.